

Translated

From: e-maxx.ru

Search the subarray with the maximum/ minimum sum

Here, we consider the problem of finding a subarray with maximum sum, as well as some of its variations (including the algorithm for solving this problem online).

Problem statement

Given an array of numbers $a[1 \dots n]$. It is required to find a subarray $a[l \dots r]$ with the maximal sum:

$$\max_{1 \leq l \leq r \leq n} \sum_{i=l}^r a[i].$$

For example, if all integers in array $a[]$ were non-negative, then the answer would be the array itself. However, the solution is non-trivial when the array can contain both positive and negative numbers.

It is clear that the problem of finding the **minimum** subarray is essentially the same, you just need to change the signs of all numbers.

Algorithm 1

Here we consider an almost obvious algorithm. (Next, we'll look at another algorithm, which is a little harder to come up with, but its implementation is even shorter.)

Algorithm description

The algorithm is very simple.

We introduce for convenience the **notation**: $s[i] = \sum_{j=1}^i a[j]$. That is, the array $s[i]$ is an array of partial sums of array $a[]$. Also, set $s[0] = 0$.

Let us now iterate over the index $r = 1 \dots n$, and learn how to quickly find the optimal l for each current value r , at which the maximum sum is reached on the subarray $[l, r]$.